

Kinematics

Frame matrix:

$$\underline{F} = \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

* $\underline{n}$: normal

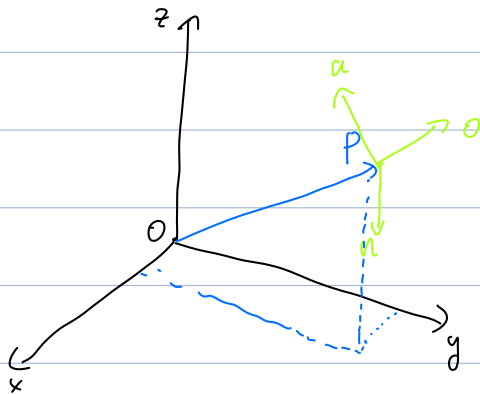
* $\underline{o}$: orientation

* $\underline{a}$: approach

$|\underline{n}| = |\underline{o}| = |\underline{a}| = 1$ unit vector.

$\underline{n}, \underline{o}, \underline{a}$: mutually perpendicular

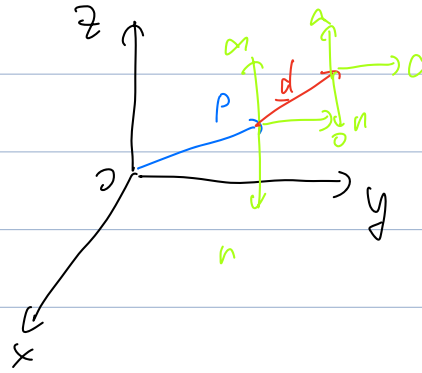
i.e. $\underline{n} \perp \underline{o} \perp \underline{a}$



Transformations

Pure translation transformation

$$T = \begin{bmatrix} 1 & 0 & 0 & d_x \\ 0 & 1 & 0 & d_y \\ 0 & 0 & 1 & d_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



$$F_{\text{new}} = \text{Trans}(dx, dy, dz) \times F_{\text{old}}$$

$$= \begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} n_x & o_x & a_x & p_x \\ n_y & o_y & a_y & p_y \\ n_z & o_z & a_z & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} n_x & o_x & a_x & p_x + dx \\ n_y & o_y & a_y & p_y + dy \\ n_z & o_z & a_z & p_z + dz \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Similarly:

$$P_{\text{new}} = \text{Trans}(dx, dy, dz) \times P_{\text{old}}$$

$$= \begin{bmatrix} p_x + dx \\ p_y + dy \\ p_z + dz \end{bmatrix}$$

Pure rotation transformation

$$R_{\text{rot}}(x, \theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & C_\theta & -S_\theta & 0 \\ 0 & S_\theta & C_\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_{\text{rot}}(y, \theta) = \begin{bmatrix} C_\theta & 0 & S_\theta & 0 \\ 0 & 1 & 0 & 0 \\ -S_\theta & 0 & C_\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_{\text{rot}}(z, \theta) = \begin{bmatrix} C_\theta & -S_\theta & 0 & 0 \\ S_\theta & C_\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Transformation notation

U_{T_A} : transformation performed on frame A
relative to fixed frame U

U: universal

$${}^U p = U_{T_A} \times {}^A p \rightarrow \text{position of frame A}$$

↓
position of frame U

Combined transformation

transformation from $V \rightarrow$ frame $A \rightarrow$ frame $B \rightarrow$ frame C

$${}^V T_A = {}^V T_C \times {}^V T_B \times {}^V T_C$$

pre-multiply

transformation from $V \rightarrow$ frame $A \rightarrow$ frame $B \rightarrow$ frame C (relative to frame A)

$${}^V T_A = {}^V T_C \times \underbrace{{}^V T_A \times {}^A T_B}$$

↓

post-multiply

↓

pre-multiply

to summarise :

* pre-multiply : transformation relative to universal frame

* post-multiply : transformation relative to current frame